WORKED SOLUTION

STEP III 2022, Question 10

Mechanics • Elastic Springs Equilibrium

STEP III 2022, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Because the angle at P is a right angle, the two spring forces are perpendicular. Resolving along each spring therefore gives each tension on its own, with no simultaneous equations. For the energy, the elastic terms combine with the gravitational term so neatly that p becomes a simple function of θ. The real content of part (ii) is then the *physical* restriction on θ: both springs must be stretched.

Solution

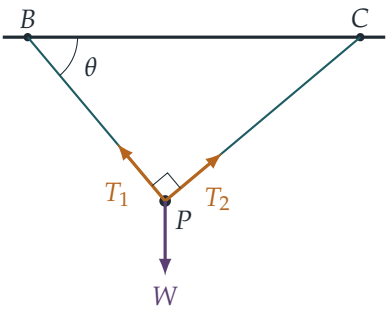
We model P as a particle and the springs as light and obeying Hooke's law, so a spring of modulus λ and natural length a stretched to length l has tension $\lambda(l-a)/a$ and stores energy $\lambda(l-a)^2/(2a)$.

Part (i)

Since $BC = 2a$ and angle $BPC = 90°$, triangle BPC has hypotenuse BC, so

$$PB = 2a\cos\theta, \qquad PC = 2a\sin\theta, \qquad \angle PCB = 90° - \theta.$$

Let T_1 and T_2 be the tensions in PB and PC.



The line PB makes angle θ with the horizontal, so it makes angle $90^\circ - \theta$ with the vertical; similarly PC makes angle θ with the vertical. As $PB \perp PC$, resolving along PB involves only T_1 and the weight, and the component of the weight along BP (away from B) is $W\cos(90^\circ - \theta) = W\sin\theta$. So

$$T_1 = W\sin\theta, \qquad \text{and likewise, along } PC, \qquad T_2 = W\cos\theta.$$

By Hooke's law,

$$T_1 = \frac{s_1 W(2a\cos\theta - a)}{a} = s_1 W(2\cos\theta - 1), \qquad T_2 = s_2 W(2\sin\theta - 1).$$

Equating,

$$s_1 = \frac{\sin\theta}{2\cos\theta - 1}, \qquad s_2 = \frac{\cos\theta}{2\sin\theta - 1},$$

which is the required s_1, together with s_2.

Part (ii)

Gravitational energy. P is below BC by the altitude of the triangle from P, which is $PB\sin\theta = 2a\sin\theta\cos\theta = a\sin 2\theta$. With zero level at BC, the gravitational potential energy is $-Wa\sin 2\theta$.

Elastic energy. The extension of PB is $a(2\cos\theta - 1)$, so, using part (i),

$$\frac{s_1 W\, a^2(2\cos\theta - 1)^2}{2a} = \tfrac{1}{2}Wa\, s_1(2\cos\theta - 1)^2 = \tfrac{1}{2}Wa\sin\theta\,(2\cos\theta - 1),$$

and similarly the energy in PC is $\frac{1}{2}Wa\cos\theta\,(2\sin\theta - 1)$.

Total. Adding,

$$\begin{aligned} -paW &= -Wa\sin 2\theta + \tfrac{1}{2}Wa(2\sin\theta\cos\theta - \sin\theta + 2\sin\theta\cos\theta - \cos\theta) \\ &= -Wa\sin 2\theta + Wa\sin 2\theta - \tfrac{1}{2}Wa(\sin\theta + \cos\theta), \end{aligned}$$

so

$$p = \tfrac{1}{2}(\sin\theta + \cos\theta) = \tfrac{\sqrt{2}}{2}\sin(\theta + \tfrac{\pi}{4}).$$

Ask yourself Can θ be anything between 0 and $\frac{\pi}{2}$?

No. The moduli $s_1 W$ and $s_2 W$ are positive, and the tensions $W\sin\theta$ and $W\cos\theta$ are positive, so both springs must be *stretched*: $2\cos\theta - 1 > 0$ and $2\sin\theta - 1 > 0$. (If either were zero, that spring would exert no force but would still need to supply a non-zero tension.) Hence $\cos\theta > \frac{1}{2}$ and $\sin\theta > \frac{1}{2}$, that is,

$$\tfrac{\pi}{6} < \theta < \tfrac{\pi}{3}, \qquad \text{so} \qquad \tfrac{5\pi}{12} < \theta + \tfrac{\pi}{4} < \tfrac{7\pi}{12}.$$

On this interval, which is symmetric about $\frac{\pi}{2}$, $\sin(\theta + \frac{\pi}{4})$ is at most 1 (attained at $\theta = \frac{\pi}{4}$) and strictly greater than its value at the endpoints, $\sin\frac{5\pi}{12} = \sin\frac{7\pi}{12}$. Now

$$\sin\tfrac{5\pi}{12} = \sin(\tfrac{\pi}{4} + \tfrac{\pi}{6}) = \tfrac{\sqrt{2}}{2}\cdot\tfrac{\sqrt{3}}{2} + \tfrac{\sqrt{2}}{2}\cdot\tfrac{1}{2} = \frac{\sqrt{2}(\sqrt{3}+1)}{4},$$

so

$$\tfrac{1}{2}\sqrt{2} \;\geq\; p \;>\; \tfrac{\sqrt{2}}{2}\cdot\frac{\sqrt{2}(\sqrt{3}+1)}{4} = \tfrac{1}{4}(1+\sqrt{3}),$$

as required.

One significant figure. We need $0.65 \leq p < 0.75$, and we justify the surd estimates by squaring. Since $1.6^2 = 2.56 < 3$, we have $\sqrt{3} > 1.6$, so $p > \frac{1}{4}(2.6) = 0.65$. Since $1.5^2 = 2.25 > 2$, we have $\sqrt{2} < 1.5$, so $p \leq \frac{1}{2}\sqrt{2} < 0.75$. Hence $p = 0.7$ to one significant figure.

Checks

- At $\theta = \frac{\pi}{4}$ the set-up is symmetric, and indeed $s_1 = s_2 = \frac{1}{\sqrt{2}-1} = \sqrt{2}+1$, with $p = \frac{1}{2}\sqrt{2} \approx 0.7071$.
- As $\theta \to \frac{\pi}{6}^+$, $s_1 \to \infty$: the spring PB is at its natural length and must be infinitely stiff to carry a tension. Then $p \to \frac{1}{4}(1+\sqrt{3}) \approx 0.6830$, the lower bound.
- We confirmed by computer, for many random θ in $(\frac{\pi}{6}, \frac{\pi}{3})$, that the position found is a stationary point of the total potential energy (so really is an equilibrium) and that the energy equals $-\frac{1}{2}Wa(\sin\theta + \cos\theta)$.

Common slips

- Giving the gravitational energy a positive sign: P is *below* the zero level.
- Maximising $\frac{1}{2}(\sin\theta + \cos\theta)$ over all of $(0, \frac{\pi}{2})$ and so finding the wrong lower bound $\frac{1}{2}$; the springs must be in tension.
- Quoting $\sqrt{3} \approx 1.732$ without showing that the error cannot change the rounding.

Beyond STEP

The cancellation in part (ii) is no accident: at an equilibrium the total potential energy is stationary, and for Hooke's-law springs the elastic energy is exactly half of (tension $\times$ extension), which ties it to the work done against gravity. Studying equilibria as stationary points of potential energy, and their stability through the second derivative, is a central theme of first-year mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops this energy viewpoint carefully.

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